GEOTECHNICAL ENGINEERING

UNIT - III : STRESS DISTRIBUTION IN SOILS & COMPACTION

Course Code: 23A01503T

Syllabus: STRESS DISTRIBUTION IN SOILS: Boussinesq's and Westergaard's theories for point loads and areas of different shapes - Newmark's influence chart. COMPACTION: Mechanism of compaction - Factors affecting - Effects of compaction on soil properties - Field compaction equipment - Compaction control.

PART A : STRESS DISTRIBUTION IN SOILS

1. Introduction

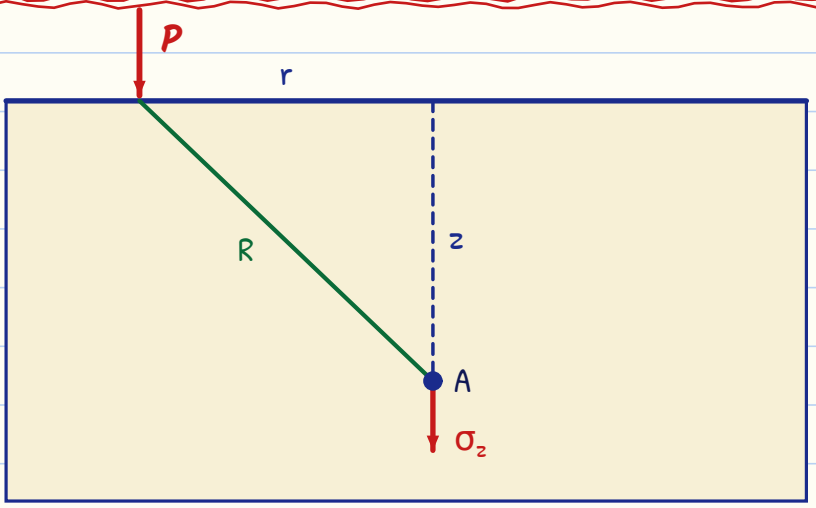
Loads from structures are transferred to the soil through foundations. The stress increase $\Delta\sigma$ at different depths below the foundation is needed to compute settlement and to check the shear failure of the soil.

- Soil is not a perfectly elastic material, but the theory of elasticity is used because it gives reasonably good results in practice (stresses are not much affected by the modulus).
- Assumptions of the elastic theories: the soil mass is homogeneous, isotropic, linearly elastic and extends to infinite depth (semi-infinite half space) with a horizontal surface; weight of the soil is ignored (only the increase in stress due to the load is computed); the loaded area is on the surface and the contact stress is known.
- Boussinesq (1885): homogeneous, isotropic, elastic half space. Westergaard (1938): elastic material reinforced by thin, rigid horizontal sheets which prevent lateral strain (stratified soils).

Types of loads

- Point (concentrated) load P. Line load q per unit length.
- Strip load (long footing, wall footing). Circular and rectangular uniformly loaded areas.
- Embankment (triangular) load, etc.

2. Boussinesq's Theory for a Point Load



Boussinesq's equation

$\sigma_z = 3P\, z^3 / (2\pi R^5)$

$= I_B\, P / z^2$

$I_B = (3/2\pi)\,[1/(1+(r/z)^2)]^{5/2}$

$R = \text{sqrt}\,(r^2 + z^2)$

max. at $r = 0$ ($I_B = 0.4775$)

For a concentrated load P acting on the surface of a semi-infinite elastic mass, the vertical stress at a point A at a depth z and radial distance r is

$$\sigma_z = (3P / 2\pi)\, z^3 / R^5 = I_B\, P / z^2$$

- $R^2 = r^2 + z^2$. The influence factor $I_B = (3 / 2\pi)\,[\,1 / (1 + (r/z)^2)\,]^{5/2}$ depends only on r/z.
- The stress is independent of the elastic constants (E and ν) in Boussinesq's equation for σ_z.
- Directly under the load ($r = 0$): $\sigma_z = 0.4775\, P / z^2$. The stress decreases rapidly with the depth and with the radial distance.

r/z	0	0.25	0.5	1.0	1.5	2.0	3.0
I_B (Boussinesq)	0.4775	0.4103	0.2733	0.0844	0.0251	0.0085	0.0015

Example 1

A point load of 500 kN acts on the surface. Find σ_z at 3 m depth and 2 m horizontal distance from the load.

- $r/z = 2/3 = 0.667$; $I_B = 0.4775 \times [1/(1 + 0.444)]^{2.5} = 0.4775 \times 0.399 = 0.190$
- $\sigma_z = 0.190 \times 500 / 3^2 = 10.6$ kPa

Other results of Boussinesq

- Line load q per unit length: $\sigma_z = 2\, q\, z^3 / [\pi\, (x^2 + z^2)^2]$.
- Uniformly loaded circular area of radius R (below the centre):

$$\sigma_z = q\,[\,1 - 1 / (1 + (R/z)^2)^{3/2}\,]$$

- Strip load of intensity q and width B: $\sigma_z = (q/\pi)\,[\,\alpha + \sin\alpha \cos(\alpha + 2\delta)\,]$, where α is the angle subtended at the point by the strip and δ is the angle between the vertical and

the line to the nearer edge.

- Rectangular area: $\sigma_z = q\, I_r$ below a corner.

Example 2 (circular footing)

q = 150 kPa, R = 2 m, z = 3 m. $(R/z)^2 = 0.444$; $1/(1.444)^{1.5} = 0.576$; $\sigma_z = 150\,(1 - 0.576) =$ 63.6 kPa.

3. Westergaard's Theory

Westergaard (1938) considered soils such as stratified deposits (varved clays, sand layers with clay seams). The soil is assumed to be an elastic medium in which thin, rigid sheets (no deformation) prevent lateral strain. For Poisson's ratio $\nu = 0$:

$$\sigma_z = (P / z^2) \times (1/\pi) / [\,1 + 2\,(r/z)^2\,]^{3/2} = I_W\, P / z^2$$

- Directly under the load: $I_W = 1/\pi = 0.3183$ (compared with $I_B = 0.4775$).

r/z	0	0.5	1.0	1.5	2.0	3.0
I_W (Westergaard)	0.3183	0.1733	0.0613	0.0247	0.0118	0.0038

Boussinesq vs Westergaard

Aspect	Boussinesq	Westergaard
Soil assumed	homogeneous, isotropic, elastic	stratified, thin rigid sheets (no lateral strain)
Depends on ν	no	yes (ν = 0 usually)
Stress below load	larger ($0.4775\ P/z^2$)	smaller ($0.3183\ P/z^2$)
Suitable for	homogeneous clay / sand	stratified, varved deposits

Boussinesq gives higher stress near the axis and is more conservative (safe). It is mostly used in practice. The Westergaard values are about 2/3 of the Boussinesq values under the load.

4. Stresses Under Areas of Different Shapes

Rectangular area (Fadum / Newmark)

Below the corner of a uniformly loaded rectangle B x L (B = width, L = length) at a depth z:

$$\sigma_z = q\, I_r \qquad I_r = f(m, n),\ m = B / z,\ n = L / z$$

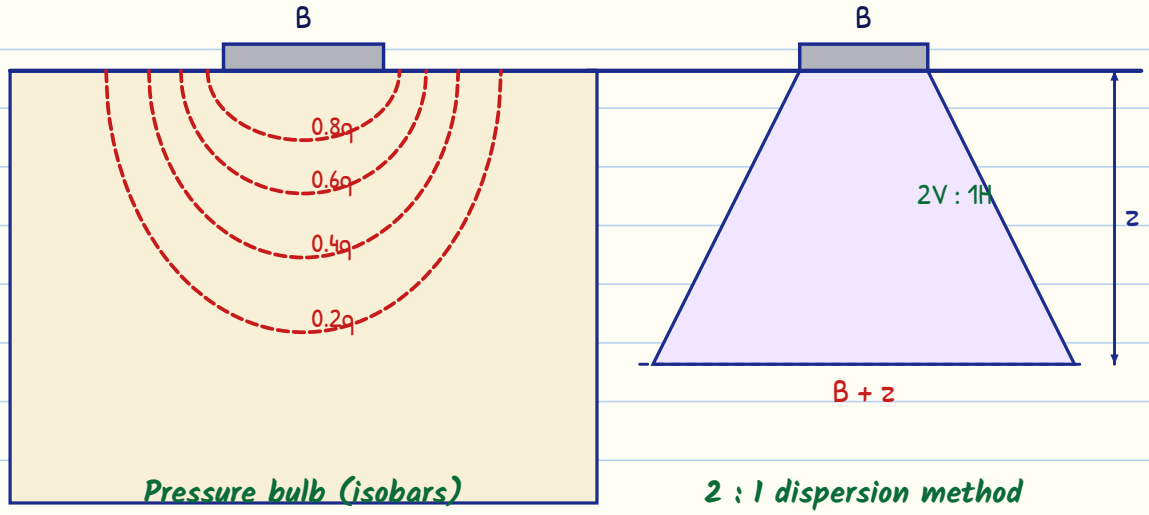
- I_r is given by charts / tables (Fadum chart); $I_r = (1/4\pi)\,[\,(2mn\,\mathrm{sqrt}(m^2+n^2+1)/(m^2+n^2+m^2n^2+1))\times((m^2+n^2+2)/(m^2+n^2+1)) + \tan^{-1}(2mn\,\mathrm{sqrt}(m^2+n^2+1)/(m^2+n^2+1-m^2n^2))\,]$
- For a point not at the corner use the superposition method: divide the area into rectangles with a common corner above the point and add (or subtract) the effects.
- Below the centre: divide into 4 equal rectangles, so $\sigma_z = 4\,q\,I_r(B/2, L/2)$.

Example 3

A 4 m x 2 m rectangular footing carries q = 200 kPa. Find σ_z at 2 m below (i) a corner, (ii) the centre.

- (i) $m = B/z = 2/2 = 1$, $n = L/z = 4/2 = 2$, $I_r = 0.1999$. $\sigma_z = 200 \times 0.1999 = 40$ kPa.
- (ii) Four rectangles of 2 m x 1 m: $m = 1/2 = 0.5$, $n = 2/2 = 1.0$, $I_r = 0.1202$. $\sigma_z = 4 \times 200 \times 0.1202 = 96.2$ kPa.

Approximate (2 : 1 distribution) method



The load is assumed to spread at a slope of 2 vertical : 1 horizontal so that the width increases by z at depth z.

Strip: $\sigma_z = q\,B / (B + z)$ Rectangle: $\sigma_z = q\,B\,L / [\,(B + z)(L + z)\,]$

- Simple, but gives the average stress; it is fairly accurate between $z = B$ and $4B$.

Example 4

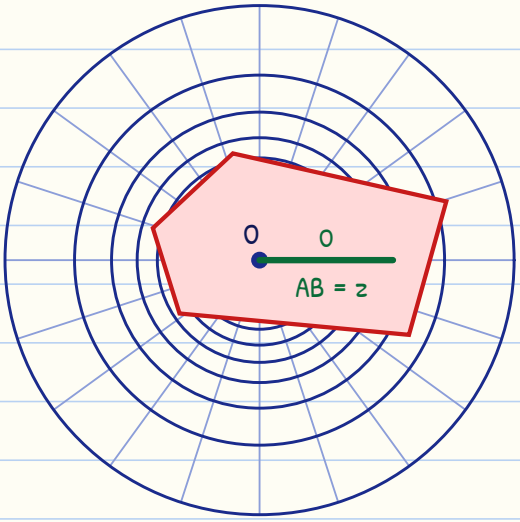
Square footing 2 m x 2 m, q = 200 kPa, find the stress at z = 3 m: $\sigma_z = 200 \times 2 \times 2 / (5 \times 5) = 32$ kPa.

Pressure bulb (isobar diagram)

- An isobar is a contour joining points of equal vertical stress below a loaded area. The pressure bulb is usually taken as the isobar of 0.2 q (20% of contact pressure) or 0.1 q.

- Significant depth (depth of influence): about 1.5 B for strip and 2 B for square footings (the bulb of 0.2 q).
- Closely spaced footings have overlapping bulbs, so the stresses add up; this is why a large foundation affects deeper soil than a small one, and settlement depends on the size of foundation.

5. Newmark's Influence Chart



Newmark chart:

9 concentric circles + 20 radial lines
(10 circles with the centre)
radii / z = 0.270, 0.401, 0.518, 0.637, 0.766, 0.918, 1.110, 1.387, 1.908

200 elements, each influence value
I = 1/200 = 0.005

AB = scale line = depth z
Footing (red) drawn to the scale AB = z with the point O below which σ_z is wanted placed at the chart centre.

Newmark (1942) developed a chart based on Boussinesq's equation for a circular loaded area, to get the vertical stress below any irregular shaped loaded area at any point.

- For a circular area of radius R carrying q, the stress ratio $\sigma_z/q = 1 - [1/(1 + (R/z)^2)]^{3/2}$. Choose σ_z/q = 0.0, 0.1, 0.2 ... 0.9, 1.0 and solve for R/z; this gives the radii of the concentric circles. (For σ_z/q = 1, R/z = infinity.)
- Then the circles are divided by 20 equally spaced radial lines, so the 10 rings x 20 = 200 elements, each having the same influence value I = 0.1 / 20 = 0.005.

Procedure

- Fix the depth z at which the stress is required and draw the plan of the loaded area to a scale such that the length of the scale line AB on the chart equals z.
- Place the plan so that the point O (vertically above which stress is required) coincides with the centre of the chart.

- Count the number of elements N covered by the plan (count partial elements by estimate).

$$\sigma_z = I \times N \times q = 0.005\, N\, q$$

- For the stress at another depth, change the scale of the plan (AB = new z). For another point, shift the plan.

Example 5

For a footing plan placed on the chart (AB = z) the number of elements covered = 62 and q = 180 kPa: σ_z = 0.005 x 62 x 180 = 55.8 kPa.

Advantages

- Can be used for loaded areas of any shape, including multiple or irregular footings, and for any point inside or outside.
- Simple to use, but a new plan has to be drawn for each depth.

PART B : COMPACTION

6. Introduction and Mechanism of Compaction

Compaction is the process of densification of soil by the application of mechanical energy (rolling, tamping, vibration), which reduces the air voids at constant water content.

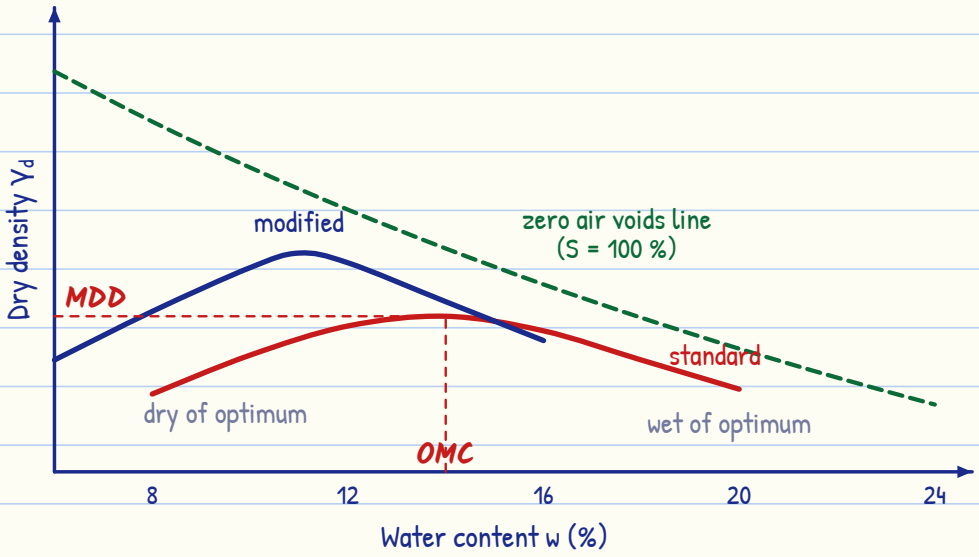
- Aims: increase shear strength, reduce compressibility (settlement), reduce permeability, reduce swelling and shrinkage and resist frost / liquefaction.
- Applications: embankments, earth dams, road subgrade and base, backfills, airfield pavements, and ground improvement.

Compaction	Consolidation
Expulsion of air by mechanical energy	Expulsion of water by static load
Short term, quick (during construction)	Long term, slow (years)
Water content constant	Water content decreases
Applies to partly saturated soil	Applies to saturated soil

Mechanism (R.R. Proctor, 1933)

- Dry soil: particles have friction and no lubrication, so air voids remain and the dry density is low.

- With a small amount of water added, water acts as a lubricant (water films) and the particles slide into closer packing, so dry density increases.
- At the optimum water content (OMC) the dry density is maximum (MDD). The air voids are at a minimum, about 5 - 10 %.
- Beyond the OMC, additional water fills the voids and keeps the grains apart; the air cannot be expelled and the pore pressure builds up, so the dry density decreases. The curve can never exceed the zero air voids line (S = 100 %).



Laboratory compaction tests (Proctor test)

Item	Standard (Light) Proctor	Modified (Heavy) Proctor
IS code	IS 2720 Part 7	IS 2720 Part 8
Mould	1000 cc (100 mm dia, 127.3 mm high)	1000 cc (same)
Rammer mass	2.6 kg	4.9 kg
Height of fall	310 mm	450 mm
Layers x blows	3 x 25	5 x 25
Energy	about 592 kJ/m^3	about 2700 kJ/m^3

- Soil passing the 19 mm / 4.75 mm sieve is mixed at 4 - 5 different water contents; each sample is compacted in the mould; bulk density γ and the water content w are found and the dry density is computed:

$$\gamma_d = \gamma / (1 + w)$$

- Plot γ_d against w to get the compaction curve. The peak gives the maximum dry density (MDD) and the optimum moisture content (OMC).
- Zero air voids line / saturation line: $\gamma_d = G \gamma_w / (1 + w G)$. For a given percentage of air voids $\gamma_d = (1 - n_a) G \gamma_w / (1 + w G)$.

Example 6

Standard Proctor test data (G = 2.70):

w (%)	10	12	14	16	18
γ bulk (g/cc)	1.89	2.01	2.08	2.06	2.00
$\gamma_d = \gamma/(1+w)$	1.718	1.795	1.825	1.776	1.695

- OMC = 14 %, MDD = 1.825 g/cc. ZAV at 14 %: 2.70/(1 + 0.14 x 2.70) = 1.959 g/cc (above the curve, as it should be).
- At OMC: e = 2.70/1.825 - 1 = 0.479; S = wG/e = 0.14 x 2.70/0.479 = 0.789 = 78.9 % (so about 21 % of the voids hold air).

7. Factors Affecting Compaction

- Water content: the most important factor - as explained above, MDD occurs only at OMC.
- Compaction effort (energy): greater effort gives a higher MDD and lower OMC (curve shifts upwards and to the left, as in the figure, modified Proctor vs standard). It is more effective on the dry side. Beyond a limit, increase in effort gives little gain.

  E = (blows x layers x rammer weight x fall height) / V

- Type of soil (gradation, grain size): well-graded coarse soils reach high MDD (2.0 - 2.3 g/cc) at low OMC (6 - 10 %); uniform sands give a flat curve; clays give lower MDD (1.4 - 1.8 g/cc) at high OMC (15 - 30 %) with a sharply peaked curve for low plasticity, a flat curve for high plastic clay.
- Method of compaction: impact (Proctor), kneading (sheepsfoot), static pressure, vibration - for sands vibration is best; for clay kneading and impact. Laboratory results differ from field results.
- Admixtures: lime, cement, bitumen and salts alter the OMC and MDD; coarse particle content (oversize) affects the results (correction for the oversize particles).
- Lift thickness and number of passes: thick layers are not compacted at the bottom.
- Temperature, initial water content, and soil structure have minor effects.

Soil type	Typical OMC (%)	Typical MDD (g/cc)
Well-graded sand / gravel	6 - 10	2.0 - 2.3
Silty sand	9 - 14	1.85 - 2.1
Silt / sandy clay	12 - 20	1.65 - 1.95
Clay (CL / CH)	20 - 30	1.4 - 1.7

Clean sand and gravel: the dry density is highest when completely dry or fully saturated (bulking in between), so compaction is controlled by relative density and flooding / vibration, rather than by OMC.

8. Effects of Compaction on Soil Properties

Property	Dry side of OMC	Wet side of OMC
Structure	flocculated (random, higher strength)	dispersed (oriented particles)
Permeability	higher	lowest at about OMC, further lower
Compressibility (low pressure)	lower (stiffer)	higher
Compressibility (high pressure)	higher (structure collapses)	lower
Shear strength (as compacted)	higher (suction)	lower
Swelling	higher (large swell on saturation)	lower
Shrinkage	lower	higher
Pore water pressure	low (negative)	high (positive)

- Increase in dry density leads to higher strength (CBR increases), lower permeability and lower settlement.
- Compacting dry of optimum gives a high strength soil, but if it later gets wet it may swell or collapse. For earth dam core and canal lining use wet of optimum (low permeability, flexible, tolerates differential settlement without cracking). For road subgrade use at or slightly dry of optimum (high CBR).
- Pore pressures and anisotropy develop during compaction in clays; the horizontal permeability is greater than the vertical (layered effect).
- Compaction increases the density and reduces the settlement of the sand and also reduces the risk of liquefaction.

9. Field Compaction Equipment

Equipment	Suitable soils / uses
Smooth-wheeled (steel drum) roller, static 8 - 10 t	sand, gravel, crushed stone, sub-base, bituminous layers (not for clay at depth)
Sheepsfoot (tamping foot) roller	cohesive soils (clays, silty clays), kneading action; foot pressure 1.5 - 7 MPa
Pneumatic tyred roller (heavy 10 - 50 t)	most soils, both sand and clay; also asphalt; adjustable tyre pressure
Vibratory roller (drum with vibrator)	granular, cohesionless soil (sand, gravel, rockfill); good for thick lifts
Vibrating plate / rammer / frog rammer	small and confined areas, trenches, near walls
Dynamic compaction (heavy tamping) 10 - 40 t weight dropped 10 - 40 m	deep compaction of loose granular fill, up to 10 m depth
Vibroflotation (vibro compaction)	clean sand deep layers, ground improvement

- Steel-wheel rollers give a smooth surface but the contact pressure is low at the bottom of the lifts.
- Sheepsfoot rollers compact from the bottom to the top, as the feet gradually "walk out" of the layer.
- Loose layer thickness: about 150 - 300 mm for rollers, 100 - 150 mm for the plate vibrators; the number of passes is generally 6 - 12 (found by a trial test strip).
- Choose the equipment from soil type, size of the job and thickness of the lift. Moisture content must be near OMC (add water by sprinkling, or dry the soil by aeration / blending).

10. Compaction Control

The aim is to ensure that the compacted fill meets the specified density and water content.

The control is done by two methods:

(a) Method specification (end product not tested)

- The type and weight of the roller, the thickness of the lift, the number of passes and the water content are specified, from a trial section (test strip).

(b) End product specification (more common)

- The relative compaction (degree of compaction) of the field-compacted soil is specified:

$$R.C. = \left(\gamma_{d,\ field} / \gamma_{d\ max,\ lab} \right) \times 100 >= 95\ \%\ \text{(often 95 - 100 \%)}$$

- Typical: embankments 95 %, earth dam core 98 - 100 % of standard Proctor MDD, road subgrade 97 % of modified Proctor.

- Water content to be maintained within OMC +/- 2 % (dry or wet side as chosen).
- For cohesionless soils (sands): control by relative density D_r >= 70 %.

Field density methods for control

- Core cutter method (clay), sand replacement method (gravel and sand), rubber balloon, nuclear density gauge (quick and widely used) and proctor needle (quick check of water content / strength).
- Frequency: one test per 1000 m^2 per layer (as in the specification) and at least 3 tests per lift.
- Check for the air void percentage (should be less than 5 - 10 %) and the saturation line: points above the ZAV line are wrong.

Example 7

Field density tests give a bulk density of 1.98 g/cc at a water content of 13 %. MDD = 1.85 g/cc.

- γ_d = 1.98/1.13 = 1.752 g/cc; R.C. = 1.752/1.85 = 0.947 = 94.7 % (below the 95 % requirement, so recompact the layer).

Problems in the field

- Over-compaction (too many passes, or too wet soil) causes "rubbering / pumping" and shearing of the layer.
- Soft spots on wet clay: dry the soil by aeration or add lime.
- Laminations between the layers: scarify the surface of the previous layer before laying the next lift.

II. Quick Revision

Item	Formula
Boussinesq point load	$\sigma_z = I_B P / z^2$, $I_B = (3/2\pi)[1/(1+(r/z)^2)]^{5/2}$
Westergaard point load	$\sigma_z = I_W P / z^2$, $I_W = (1/\pi)/[1+2(r/z)^2]^{3/2}$
Circular area (centre)	$\sigma_z = q\,[1 - (1/(1+(R/z)^2))^{3/2}]$
Rectangle (corner)	$\sigma_z = q\, I_r(m, n)$
2:1 method	$\sigma_z = q\, B\, L / [(B+z)(L+z)]$
Newmark chart	$\sigma_z = 0.005\, N\, q$
Dry density	$\gamma_d = \gamma/(1+w)$

ZAV line	$\gamma_d = G \gamma_w / (1 + w G)$
Relative compaction	R.C. = $\gamma_{d\ field} / \gamma_{d\ max}$

Important questions (exam)

- Derive / state Boussinesq's equation for vertical stress below a point load. How does Westergaard's analysis differ?
- Explain how stress beneath a rectangular area is found. What is the 2:1 method?
- What is Newmark's influence chart? Explain its construction and use.
- Explain the mechanism of compaction. Explain the standard Proctor test and the compaction curve.
- Discuss the factors that affect compaction and the effects of compaction on soil properties.
- Describe the field compaction equipment and the methods of compaction control.